

ELEC4631 Continuous-Time Control System Design

Lectures 1 and 2: Brief Review of Linear Algebra and Vector Calculus

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1 Introduction

The first two lectures provide a brief review of concepts from linear algebra and vector calculus that you have already encountered in the maths courses which you have taken. The notes only serve to give you a reminder of those concepts and do not go into them in detail. If you feel you need to, you should pick up your maths textbooks and do a more in-depth review in your own time.

2 Eigenvalues and eigenvectors

Recall that a complex number λ is an eigenvalue of a $n \times n$ (real or complex) square matrix A if there exists a complex $n \times 1$ non-zero column vector v ($v \neq 0$) such that $Av = \lambda v$. We make the convention throughout that any element of $\mathbb{R}^n$ and $\mathbb{C}^n$ will be represented as a column vector of real and complex numbers, respectively. The matrix A has n eigenvalues (some possibly repeated) and they can all be found by solving the *characteristic equation* $\det(\lambda I_{n \times n} - A) = 0$. When the dimension is clear from the context we shall write the $n \times n$ identity matrix $I_{n \times n}$ simply as I . The left hand side of the equation is an n -th order polynomial in λ and the n roots of this polynomial are the eigenvalues of A . The polynomial $c_A(\lambda) = \det(\lambda I - A)$ is referred to as the *characteristic polynomial* of A .

Let $\mathbf{a}_\lambda$ denote the *algebraic multiplicity* of λ as a root of $c_A(\lambda)$. That is, how many times λ appears as a root of the characteristic polynomial. For instance, if $c_A(\lambda) = (\lambda - 4)^3(\lambda + 1)$ then the algebraic multiplicity of the eigenvalue $\lambda = 4$ is 3 (i.e., $\mathbf{a}_4 = 3$) while the algebraic multiplicity of the eigenvalue -1 is 1 (i.e., $\mathbf{a}_{-1} = 1$).

Example 1 *The matrix*

$$A = \begin{bmatrix} 1 & 2 \\ -1 & 4 \end{bmatrix}$$

has characteristic polynomial $c_A(\lambda) = (\lambda - 3)(\lambda - 2)$, so its eigenvalues are 3 and 2, each of multiplicity 1.

Example 2 *The matrix*

$$A = \begin{bmatrix} 5 & -1 & 0 \\ 0 & -4 & 1 \\ 0 & -1 & -2 \end{bmatrix}$$

has characteristic polynomial $c_A(\lambda) = (\lambda - 5)(\lambda + 3)^2$, so its eigenvalues are 5, -3 and -3. The eigenvalue 5 is of multiplicity 1 and the eigenvalue -3 has multiplicity 2.

Note the following properties. If A is a real matrix (its entries are all real) then $c_A(\lambda)$ will have real coefficients. This then implies that all non-real eigenvalues of A always come in conjugate pairs. That is, if A is real and $\lambda = x + jy$ (here $j = \sqrt{-1}$) is a non-real eigenvalue then $\lambda^* = x - jy$ (the complex conjugate of λ) is also an eigenvalue of A and λ^* has the same algebraic multiplicity as λ . Once an eigenvalue λ has been determined, a corresponding eigenvector (not unique) or eigenvectors (there can be more than one linearly independent eigenvectors associated to a single eigenvalue) can be found by solving the *eigenvalue equation*

$$(A - \lambda I)v = 0 \Leftrightarrow Av = \lambda v,$$

for $v \neq 0$. In other words, for a given eigenvalue λ , v is a non-zero element of the *null space* (also known as the *kernel space*) of $A - \lambda I$. We denote this null space by $\text{null}(A - \lambda I)$ and it is defined as $\text{null}(A - \lambda I) = \{z \in \mathbb{C}^n \mid (A - \lambda I)z = 0\}$. Now, being a vector space, $\text{null}(A - \lambda I)$ has a dimension. The dimension of this null space is referred to as the *geometric multiplicity* of the eigenvalue λ and will be denoted by $\mathbf{g}_\lambda$. The geometric multiplicity $\mathbf{g}_\lambda$ and the algebraic multiplicity $\mathbf{a}_\lambda$ of an eigenvalue λ *need not coincide*. Note also that we always have that $1 \leq \mathbf{g}_\lambda \leq \mathbf{a}_\lambda$. Also, eigenvectors corresponding to distinct eigenvalues are linearly independent. That is, if v_1 is an eigenvector corresponding to the eigenvalue λ_1 , and v_2 is an eigenvector corresponding to the eigenvalue λ_2 , with $\lambda_1 \neq \lambda_2$, then $c_1 v_1 + c_2 v_2 = 0$ implies that $c_1 = c_2 = 0$. Recall that a collection of vectors $v_1, v_2, \dots, v_n$ are linearly independent if $a_1 v_1 + a_2 v_2 + \dots + a_n v_n = 0$ implies that $a_1 = a_2 = \dots = a_n = 0$.

Example 3 *For the matrix*

$$\begin{bmatrix} 1 & 2 \\ -1 & 4 \end{bmatrix},$$

from an earlier example, the eigenvalue equation for $\lambda = 3$ is

$$\begin{bmatrix} -2 & 2 \\ -1 & 1 \end{bmatrix} v = 0,$$

which has $v = (1, 1)^T$ as a solution. This is an eigenvector corresponding to the eigenvalue $\lambda = 3$. The eigenvalue equation for $\lambda = 2$ is

$$\begin{bmatrix} -1 & 2 \\ -1 & 2 \end{bmatrix} v = 0,$$

which has $v = (2, 1)^T$ as a solution. This is an eigenvector corresponding to the eigenvalue $\lambda = 2$.

Example 4 For the matrix

$$\begin{bmatrix} 5 & -1 & 0 \\ 0 & -4 & 1 \\ 0 & -1 & -2 \end{bmatrix},$$

the eigenvalue equation for $\lambda = 5$ is

$$\begin{bmatrix} 0 & -1 & 0 \\ 0 & -9 & 1 \\ 0 & -1 & -7 \end{bmatrix} v = 0,$$

which has $v_1 = (1, 0, 0)^T$ as a solution. This is an eigenvector corresponding to the eigenvalue $\lambda = 5$. The eigenvalue equation for the eigenvalue $\lambda = -3$ is:

$$\begin{bmatrix} 8 & -1 & 0 \\ 0 & -1 & 1 \\ 0 & -1 & 1 \end{bmatrix} v = 0,$$

which has $v_2 = (1, 8, 8)^T$ as a solution. This is an eigenvector corresponding to the eigenvalue $\lambda = -3$. In fact, you can check that there will be no other eigenvectors associated with $\lambda = -3$ that will be linearly independent of v_2 . So, here the null space of $A + 3I$ has dimension 1. That is, it has only one basis vector, which can be chosen to be v_2 . Therefore the geometric multiplicity of the eigenvalue -3 is 1.

Example 5 The matrix

$$\begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix},$$

has complex eigenvalues $1 + j$ and $1 - j$.

The eigenvalue equation for the eigenvalue $\lambda = 1 + j$ is

$$\begin{bmatrix} -j & 1 \\ -1 & -j \end{bmatrix} v = 0,$$

which has $v = (1, j)^T$ as a solution. This is a complex eigenvector corresponding to the eigenvalue $\lambda = 1 + j$.

The eigenvalue equation for the eigenvalue $\lambda = 1 - j$ is

$$\begin{bmatrix} j & 1 \\ -1 & j \end{bmatrix} v = 0,$$

which has $v = (1, -j)^T$ as a solution. This is a complex eigenvector corresponding to the eigenvalue $\lambda = 1 - j$.

3 Positive definite and positive semidefinite matrices

A real symmetric $n \times n$ matrix A is said to be

- *Positive definite* if $x^T Ax > 0$ for all real vectors $x \neq 0$ in $\mathbb{R}^n$.
- *Positive semidefinite* if $x^T Ax \geq 0$ for all x in $\mathbb{R}^n$.

The above properties can be easily determined by inspecting eigenvalues:

- A is positive definite if and only if all its eigenvalues are positive (> 0),
- A is positive semidefinite if and only if all its eigenvalues are nonnegative (≥ 0).

Note that by definition all positive definite matrices are also positive semidefinite, but positive semidefinite matrices need not be positive definite. In a similar way, a matrix is said to be negative definite (resp., negative semidefinite) if $-A$ is positive definite (resp., positive semidefinite).

Example 6 Consider the symmetric matrix $A = \begin{bmatrix} 2 & 1 \\ 1 & 4 \end{bmatrix}$. This matrix has eigenvalues $3 \pm \sqrt{2} > 0$. Therefore it is a positive definite matrix.

Example 7 Consider the symmetric matrix $A = \begin{bmatrix} -2 & -2 \\ -2 & -2 \end{bmatrix}$. This matrix has eigenvalues -4 and 0 . Therefore it is a negative semidefinite matrix since all its eigenvalues are nonpositive (≤ 0) or, equivalently, all the eigenvalues of $-A$ are nonnegative.

4 Similarity transformations

Two square matrices A and B of dimension $n \times n$ are said to be *similar* if there exists an invertible $n \times n$ matrix V such that $A = VB V^{-1}$. In this case, the transformation from A to B (and vice-versa) is said to be a *similarity transformation*.

Similarity transformations arise from *coordinate transformations*. Suppose that a coordinate system in $\mathbb{R}^n$ has the basis vectors $\{b_1, b_2, \dots, b_n\}$ (these vectors are linearly independent but they need not be mutually orthogonal). Any point x in $\mathbb{R}^n$ can be written as $x = \alpha_1(x)b_1 + \alpha_2(x)b_2 + \dots + \alpha_n(x)b_n$ and can be represented with respect to this coordinate system as the coordinate vector $\alpha(x) = (\alpha_1(x), \alpha_2(x), \dots, \alpha_n(x))^T$. Suppose that T is a linear transformation from $\mathbb{R}^n$ to $\mathbb{R}^n$. Let $y = T(x)$ and let $\alpha(y)$ be the coordinate of y in the given coordinate system. This simply means that if $\alpha(y) = (\alpha_1(y), \alpha_2(y), \dots, \alpha_n(y))^T$ then we have that $y = \alpha_1(y)b_1 + \alpha_2(y)b_2 + \dots + \alpha_n(y)b_n$. Then there is a real square matrix A such that $\alpha(y) = A\alpha(x)$. This matrix A is a *representation* of the transformation T with

respect to the coordinate system with basis vectors $\{b_1, b_2, \dots, b_n\}$. In fact, A is given by

$$A = \begin{bmatrix} \alpha_1(T(b_1)) & \alpha_1(T(b_2)) & \dots & \alpha_1(T(b_n)) \\ \alpha_2(T(b_1)) & \alpha_2(T(b_2)) & \dots & \alpha_2(T(b_n)) \\ \vdots & \ddots & \ddots & \vdots \\ \alpha_n(T(b_1)) & \alpha_n(T(b_2)) & \dots & \alpha_n(T(b_n)) \end{bmatrix}.$$

Now, suppose that we want to change to a coordinate system in $\mathbb{R}^n$ with a new set of basis vectors $\{c_1, c_2, \dots, c_n\}$. Suppose also that in this new coordinate system the old coordinate basis $b_1, b_2, \dots, b_n$ have the coordinate vectors $v_1, v_2, \dots, v_n$, respectively. This simply means that if $v_j = (v_{j1}, v_{j2}, \dots, v_{jn})^T$ then $b_j = v_{j1}c_1 + v_{j2}c_2 + \dots + v_{jn}c_n$ for $j = 1, 2, \dots, n$. Thus, in the new coordinate system, the points x and y are now represented by the new coordinates $\alpha'(x) = V\alpha(x)$ and $\alpha'(y) = V\alpha(y)$, respectively, where

$$V = \begin{bmatrix} v_1 & v_2 & \dots & v_n \end{bmatrix}.$$

That is, in the new coordinate system, the relation between $\alpha(y) = A\alpha(x)$ in the old coordinate system becomes the relation $V^{-1}\alpha'(y) = AV^{-1}\alpha'(x) \Leftrightarrow \alpha'(y) = VAV^{-1}\alpha'(x)$. Therefore, the transformation represented by the matrix A in the old coordinate system becomes represented by the matrix $A' = VAV^{-1}$ in the new coordinate system.

Remark 8 When the basis is not specified, the usual convention is that a given $n \times n$ matrix A is a representation of some linear map T with respect to the standard basis vectors $e_1 = (1, 0, 0, \dots, 0)^T$, $e_2 = (0, 1, 0, \dots, 0)^T$, $\dots$, $e_n = (0, 0, \dots, 0, 1)^T$.

Example 9 Suppose that the vector space is $\mathbb{R}^2$ and we have as basis some linearly independent vectors f_1 and f_2 . Suppose that T is a linear transformation which maps f_1 to $T(f_1) = 3f_1 - 2f_2$ and f_2 to $T(f_2) = -2f_1 + f_2$. Now any point x in $\mathbb{R}^2$ can be expressed as $x = \alpha_1(x)f_1 + \alpha_2(x)f_2$, so that x has the coordinate $(\alpha_1(x), \alpha_2(x))^T$ with respect to the given basis. The transformation T applied to x gives

$$\begin{aligned} y &= T(x) \\ &= T(\alpha_1(x)f_1 + \alpha_2(x)f_2), \\ &= \alpha_1(x)T(f_1) + \alpha_2(x)T(f_2), \\ &= \alpha_1(x)(3f_1 - 2f_2) + \alpha_2(x)(-2f_1 + f_2), \\ &= (3\alpha_1(x) - 2\alpha_2(x))f_1 + (-2\alpha_1(x) + \alpha_2(x))f_2. \end{aligned}$$

So, T maps a point x with coordinate $(\alpha_1(x), \alpha_2(x))^T$ to another point y with coordinates $(\alpha_1(y), \alpha_2(y))^T = (3\alpha_1(x) - 2\alpha_2(x), -2\alpha_1(x) + \alpha_2(x))^T$. Now, you see that

$$\begin{bmatrix} \alpha_1(y) \\ \alpha_2(y) \end{bmatrix} = \begin{bmatrix} 3 & -2 \\ -2 & 1 \end{bmatrix} \begin{bmatrix} \alpha_1(x) \\ \alpha_2(x) \end{bmatrix},$$

so that in the given basis, the linear map T is represented by the matrix

$$A = \begin{bmatrix} 3 & -2 \\ -2 & 1 \end{bmatrix}.$$

The matrix representation will of course change if the basis is changed, as we will now see.

Suppose that we want to change basis to a new basis given by the standard basis vectors $e_1 = (1, 0)^T$ and $e_2 = (0, 1)^T$ and we are told that $f_1 = e_1 + e_2$ and $f_2 = e_1 - e_2$. What does the matrix representation of T now look like with respect to this new basis? Well, first we have that f_1 and f_2 have coordinates $f_1 = (1, 1)^T$ and $f_2 = (1, -1)^T$ with respect to this basis, so the matrix V here is

$$V = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}.$$

This means that the matrix representation of T in the standard basis vectors, which we denote by A' , is:

$$\begin{aligned} A' &= VAV^{-1}, \\ &= \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 3 & -2 \\ -2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}^{-1}, \\ &= \begin{bmatrix} 0 & 1 \\ 1 & 4 \end{bmatrix}. \end{aligned}$$

5 Diagonalisability and diagonalising a matrix

Suppose that the $n \times n$ matrix A has m distinct eigenvalues ($1 \leq m \leq n$) $\lambda_1, \dots, \lambda_m$ and each eigenvalue has algebraic multiplicity *equal* to its geometric multiplicity: $\mathbf{a}_{\lambda_k} = \mathbf{g}_{\lambda_k}$ for $k = 1, \dots, m$. Note that $\mathbf{a}_{\lambda_1} + \dots + \mathbf{a}_{\lambda_m} = n$ and thus also $\mathbf{g}_{\lambda_1} + \dots + \mathbf{g}_{\lambda_m} = n$. Therefore for each distinct eigenvalue λ_k we have $\mathbf{a}_{\lambda_k}$ linearly independent eigenvectors $v_{k1}, v_{k2}, \dots, v_{k\mathbf{a}_{\lambda_k}} \in \text{null}(A - \lambda_k I)$. All of the eigenvectors from the distinct eigenvalues will also be linearly independent of one another. Thus we have a collection of n linearly independent vectors $v_{11}, v_{12}, \dots, v_{1\mathbf{a}_{\lambda_1}}, v_{21}, v_{22}, \dots, v_{2\mathbf{a}_{\lambda_2}}, \dots, v_{m1}, v_{m2}, \dots, v_{m\mathbf{a}_{\lambda_m}}$. From the entire collection of eigenvectors we can form an $n \times n$ matrix V defined as:

$$V = \begin{bmatrix} v_{11} & v_{12} & \dots & v_{1\mathbf{a}_{\lambda_1}} & v_{21} & v_{22} & \dots & v_{2\mathbf{a}_{\lambda_2}} & \dots & v_{m1} & v_{m2} & \dots & v_{m\mathbf{a}_{\lambda_m}} \end{bmatrix}.$$

Since all the columns of V are linearly independent of one another, V is therefore an invertible matrix possessing an inverse V^{-1} .

Now, we have the relation $Av_{kl} = \lambda_k v_{kl}$ for $l = 1, \dots, \mathbf{a}_{\lambda_k}$ and for each $k = 1, \dots, m$. Using the matrix V , these relations can be compactly expressed as (you may verify this yourself):

$$AV = VD,$$

where D is an $n \times n$ diagonal matrix given by $D = \text{diag}(\underbrace{\lambda_1, \dots, \lambda_1}_{\mathbf{a}_{\lambda_1} \text{ times}}, \underbrace{\lambda_2, \dots, \lambda_2}_{\mathbf{a}_{\lambda_2} \text{ times}}, \dots, \underbrace{\lambda_m, \dots, \lambda_m}_{\mathbf{a}_{\lambda_m} \text{ times}})$.

Here, the notation $\text{diag}(c_1, c_2, \dots, c_n)$ denotes the diagonal or block diagonal matrix

$$\text{diag}(c_1, c_2, \dots, c_n) = \begin{bmatrix} c_1 & 0 & 0 & \dots & 0 \\ 0 & c_2 & 0 & \dots & 0 \\ 0 & 0 & c_3 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 0 & c_n \end{bmatrix},$$

where this matrix is diagonal if $c_1, c_2, \dots, c_n$ are scalars and is block diagonal if they are square matrices (not necessarily of the same dimensions). Since V is invertible, it follows that $D = V^{-1}AV$. That is, under a similarity transformation by V^{-1} , the matrix A is transformed into the diagonal matrix D . We say in this case that the matrix A is *diagonalisable*. Note that in this definition the transformation must be a similarity transformation.

The essence of this section is that if all the distinct eigenvalues of A have algebraic multiplicity equal to its geometric multiplicity then A is diagonalisable and this diagonalisation can be performed with the procedure described above. Conversely, for a matrix to be diagonalisable it is *necessary* for all its distinct eigenvalues to have geometric multiplicity equal to its algebraic multiplicity (as the construction above shows). However, not all matrices will have this property and thus not all matrices are diagonalisable. On the other hand, there are certain special classes of matrices that can be shown to always be diagonalisable. This includes the class of all *normal matrices* that commute with its conjugate transpose, $AA^* = A^*A$, where $*$ denotes the conjugate transpose. Special cases of normal matrices are real symmetric matrices (real matrices with the property that $A = A^T$) and complex Hermitian matrices (all complex matrices with the property $A = A^*$). Moreover,

1. For a real and symmetric matrix A we also have the additional property that all its eigenvalues are real and V can be chosen to be real, and $V^{-1} = V^T$ (V is an orthogonal matrix) provided that all eigenvectors (in the columns of V) are normalised to have norm 1.
2. Similarly, for a complex Hermitian matrix A we have that all its eigenvalues are real (but V is complex) and $V^{-1} = V^*$ (i.e., V is unitary) provided that all eigenvectors (in the columns of V) are normalised to have norm 1.

Example 10 Consider the 2×2 matrix

$$A = \begin{bmatrix} 16 & -30 \\ 9 & -17 \end{bmatrix}.$$

We can compute the characteristic polynomial $c_A(\lambda) = \det(\lambda I - A) = \lambda^2 + \lambda - 2$. Thus the eigenvalues of the matrix are 1 and -2. In this case A is diagonalisable because there are 2 distinct eigenvalues and here each must have geometric multiplicity 1.

To find the eigenvector corresponding to $\lambda = 1$ we need to solve the equation

$$(A - I)v = \begin{bmatrix} 15 & -30 \\ 9 & -18 \end{bmatrix} v = 0,$$

which has $v = c(2, 1)^T$ as a family of solutions for any (real or complex) number $c \neq 0$. Thus all such vectors are eigenvectors of A , here we will choose $v_1 = (2, 1)^T$. To find the eigenvector corresponding to $\lambda = -2$ we need to solve the equation

$$(A + 2I)v = \begin{bmatrix} 18 & -30 \\ 9 & -15 \end{bmatrix} v = 0,$$

which has $v = c(1\frac{2}{3}, 1)^T$ as a family of solutions. We can choose $v_2 = (5, 3)^T$ as an eigenvector of A .

From the chosen eigenvectors, we can construct the matrix $V = [v_1 \ v_2] = \begin{bmatrix} 2 & 5 \\ 1 & 3 \end{bmatrix}$. Since $AV = V\text{diag}(1, -2)$, V diagonalises A : $V^{-1}AV = \text{diag}(1, -2)$.

Example 11 Consider the 2×2 matrix

$$A = \begin{bmatrix} -13 & 25 \\ -4 & 7 \end{bmatrix}.$$

We can compute the characteristic function $c_A(\lambda) = \det(\lambda I - A) = (\lambda + 3)^2$. Thus A only has a single eigenvalue of algebraic multiplicity 2. Since it is impossible for this eigenvalue to have geometric multiplicity 2 (the dimension of the null space of a non-zero $n \times n$ matrix cannot be higher than $n - 1$), A cannot be diagonalised.

If a matrix A is diagonalisable and λ is an eigenvalue of algebraic multiplicity $\mathfrak{a}_\lambda > 1$ then one can always choose $\mathfrak{a}_\lambda$ linearly independent eigenvectors $v_1, \dots, v_{\mathfrak{a}_\lambda}$ in $\text{null}(A - \lambda I)$. In particular, when A is real symmetric these eigenvectors can be chosen to be orthonormal, $v_j^T v_k = 0$ if $j \neq k$ and $\|v_j\|^2 = v_j^T v_j = 1$ for $j = 1, \dots, \mathfrak{a}_\lambda$. Similarly, if A is a complex Hermitian matrix the eigenvectors can be chosen to be orthonormal, $v_j^* v_k = 0$ if $j \neq k$ and $\|v_j\|^2 = v_j^* v_j = 1$ (recall that v_j^* is the conjugate transpose of v_j).

Example 12 Consider the real symmetric matrix

$$A = \begin{bmatrix} 1/2 & -1/2 & 1/\sqrt{2} \\ -1/2 & 1/2 & 1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} & 0 \end{bmatrix}.$$

It has two eigenvalues $\lambda_1 = 1$ and $\lambda_2 = -1$ with $\mathfrak{a}_{\lambda_1} = 2$ and $\mathfrak{a}_{\lambda_2} = 1$. Two orthonormal elements of $\text{null}(A - \lambda_1 I) = \text{null}(A - I)$ are $v_{11} = (1/\sqrt{2}, -1/\sqrt{2}, 0)^T$ and $v_{12} = (1/2, 1/2, 1/\sqrt{2})^T$. A normalised eigenvector corresponding to λ_2 is $v_2 = (-1/2, -1/2, 1/\sqrt{2})^T$. Therefore, an orthogonal matrix V that diagonalises A is:

$$V = \begin{bmatrix} 1/\sqrt{2} & 1/2 & -1/2 \\ -1/\sqrt{2} & 1/2 & -1/2 \\ 0 & 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix}.$$

6 Quadratic forms

Let $x = (x_1, x_2, \dots, x_n)^T$ denote a vector in $\mathbb{R}^n$. A quadratic form on $\mathbb{R}^n$ is a function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ of x that has the form:

$$f(x) = \sum_{i,j=1}^n a_{ij}x_i x_j,$$

with real coefficients a_{ij} . For example, if $n = 3$ then $f(x) = x_1^2 + 4x_1x_2 - 2x_2x_3 + x_2^2 - x_3^2$ is a quadratic form.

Any quadratic form can be expressed in the form $f(x) = x^T A x$ where A is a real symmetric matrix of the form $A = [\frac{1}{2}(a_{ij} + a_{ji})]_{i,j=1,2,\dots,n}$. Equivalently, using the Euclidean inner product between vectors it can be expressed as $f(x) = \langle x, Ax \rangle$, where $\langle \cdot, \cdot \rangle$ is the Euclidean inner product $\langle x, y \rangle = x^T y$ on $\mathbb{R}^n$. A quadratic form is said to be

- *Positive definite* if $f(x) > 0$ for all $x \neq 0$.
- *Positive semidefinite* if $f(x) \geq 0$ for all x .

These properties are easy to verify:

- f is positive definite if and only if A is positive definite.
- f is positive semidefinite if and only if A is positive semidefinite.

Analogously, f is a negative definite (resp., negative semidefinite) quadratic form if $-f$ is positive definite (resp., positive semidefinite).

Since A is symmetric it is diagonalisable and there exists a real diagonal matrix $D = \text{diag}(d_1, d_2, \dots, d_n)$ and a real invertible matrix V such that $A = V D V^T$ (recall that for a symmetric matrix, its eigenvalues are real, and V can be chosen to be real and orthogonal: $V^{-1} = V^T$). If we define $z = V^T x$, which is equivalent to $x = Vz$, we can make a change of variable from x to z and the quadratic form $f(x)$ can be written as

$$f(x) = x^T A x = (Vz)^T A V z = z^T (V^T A V) z = z^T D z.$$

In the new variable z , the quadratic form $f(x)$ becomes the simpler quadratic form $g(z)$ given by

$$g(z) = f(Vz) = d_1 z_1^2 + d_2 z_2^2 + \dots + d_n z_n^2,$$

note that $g(z)$ only involves squares of the z_j 's and no cross-products like $z_2 z_3$.

Example 13 The quadratic form $f(x) = 8x_1^2 - 4x_1x_2 + 5x_2^2$ with $x = (x_1, x_2)$ can be written as:

$$f(x) = x^T \begin{bmatrix} 8 & -2 \\ -2 & 5 \end{bmatrix} x.$$

The eigenvalues of the matrix

$$A = \begin{bmatrix} 8 & -2 \\ -2 & 5 \end{bmatrix}$$

are 4 and 9, so this quadratic form is positive definite, i.e., the quadratic form can only take on positive values except at $x = 0$, where it takes on the value 0. The matrix A can be diagonalised by the matrix

$$V = \frac{1}{\sqrt{5}} \begin{bmatrix} 1 & -2 \\ 2 & 1 \end{bmatrix}.$$

Taking $z = V^T x$ then $f(x)$ can be transformed into the quadratic form

$$g(z) = f(Vz) = 4z_1^2 + 9z_2^2.$$

Example 14 The quadratic form $f(x) = x_1^2 + 4x_1x_2 - 2x_2x_3 + x_2^2 - x_3^2$ with $x = (x_1, x_2, x_3)$ can be written as:

$$f(x) = x^T \begin{bmatrix} 1 & 2 & 0 \\ 2 & 1 & -1 \\ 0 & -1 & -1 \end{bmatrix} x.$$

The eigenvalues of the matrix

$$\begin{bmatrix} 1 & 2 & 0 \\ 2 & 1 & -1 \\ 0 & -1 & -1 \end{bmatrix}$$

are -1.7616, -0.3633, and 3.1249, so this quadratic form is indefinite, i.e., the form can take on positive and negative values. Using the matrix

$$V = \begin{bmatrix} 0.4018 & -0.6189 & 0.6749 \\ -0.5548 & 0.4219 & 0.7171 \\ -0.7285 & -0.6626 & -0.1738 \end{bmatrix},$$

and taking $z = V^T x$ the quadratic $f(x)$ can be transformed into the quadratic form

$$g(z) = f(Vz) = -1.7616z_1^2 - 0.3633z_2^2 + 3.1249z_3^2.$$

7 The Jordan canonical form

So, what can we do if a matrix is not diagonalisable? Is there a different simpler form that it can be transformed into? Yes, there is and the form was discovered by French mathematician Camille Jordan. This is the subject of this section.

Let's begin by defining a special matrix structure called a *Jordan block*. A Jordan block is simply a square matrix of the form

$$\begin{bmatrix} c & 1 & 0 & 0 & \dots & 0 \\ 0 & c & 1 & 0 & \dots & 0 \\ 0 & 0 & c & 1 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & c & 1 \\ 0 & 0 & 0 & \dots & 0 & c \end{bmatrix}$$

for some number c (real or complex). **Note:** We can think of a scalar c as a 1×1 Jordan block. That is, $c \equiv [c]$.

Suppose that an $n \times n$ square matrix A has distinct eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_m, \kappa_1, \kappa_2, \dots, \kappa_r$ with all λ_j having algebraic multiplicity 1 and all κ_j having algebraic multiplicity $\mathbf{a}_{\kappa_j} > 1$. Then $m + \mathbf{a}_{\kappa_1} + \mathbf{a}_{\kappa_2} + \dots + \mathbf{a}_{\kappa_r} = n$.

Let's now look at a particular repeated eigenvalue κ_s with algebraic multiplicity $\mathbf{a}_{\kappa_s}$ and geometric multiplicity $\mathbf{g}_{\kappa_s}$. Thus the dimension of the null space $\text{null}(A - \kappa_s I)$ is $\mathbf{g}_{\kappa_s}$. Let $J_{\kappa_s, k}$ denote a Jordan block of some unspecified dimension for $k = 1, 2, \dots, \mathbf{g}_{\kappa_s}$ (i.e., there are as many Jordan blocks as the geometric multiplicity of κ_s) such that

$$\dim(J_{\kappa_s, 1}) + \dim(J_{\kappa_s, 2}) + \dots + \dim(J_{\kappa_s, \mathbf{g}_{\kappa_s}}) = \mathbf{a}_{\kappa_s},$$

where $\dim(X)$ denotes the row or column size of a square matrix X . That is, the dimensions of all the Jordan blocks $J_{\kappa_s, k}$ should sum to the algebraic multiplicity of κ_s —keep in mind that different $J_{\kappa_s, k}$'s can be of different dimensions. Now, for each κ_s define the block diagonal matrix

$$J_{\kappa_s} = \begin{bmatrix} J_{\kappa_s, 1} & 0 & 0 & \dots & 0 \\ 0 & J_{\kappa_s, 2} & 0 & \dots & 0 \\ 0 & 0 & J_{\kappa_s, 3} & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & \dots & 0 & \dots & J_{\kappa_s, \mathbf{g}_{\kappa_s}} \end{bmatrix}.$$

That is, this $\mathbf{a}_{\kappa_s} \times \mathbf{a}_{\kappa_s}$ matrix is zero everywhere except on the diagonal blocks that are occupied by $J_{\kappa_s, 1}, \dots, J_{\kappa_s, \mathbf{g}_{\kappa_s}}$. A (block diagonal) matrix that has Jordan blocks on its diagonal blocks and zero everywhere else is said to be in *Jordan canonical form* or *Jordan normal form*. Thus the matrix J_{κ_s} above is an example of a matrix that is in Jordan canonical form. Now, let's note the following points regarding two extreme scenarios:

1. If for a particular repeating eigenvalue κ_s we have that $\mathbf{a}_{\kappa_s} = \mathbf{g}_{\kappa_s}$ (algebraic multiplicity is equal to geometric multiplicity) then all Jordan blocks $J_{\kappa_s, k}$ ($k = 1, \dots, \mathbf{g}_{\kappa_s}$) will be 1×1 . Moreover, if this is true for all repeated eigenvalues κ_s ($s = 1, \dots, r$) then it follows that A is diagonalisable (as we had seen earlier).
2. If $\mathbf{g}_{\kappa_s} = 1$ then there will only be a single Jordan block of size $\mathbf{a}_{\kappa_s} \times \mathbf{a}_{\kappa_s}$. That is, $J_{\kappa_s} = J_{\kappa_s, 1}$.

Having defined the Jordan canonical form we can now state the following theorem:

Theorem 15 (Camille Jordan) *There exists an invertible $n \times n$ (possibly complex) ma-*

trix V that transforms the $n \times n$ matrix A into Jordan canonical form. That is,

$$V^{-1}AV = \begin{bmatrix} \lambda_1 & 0 & \dots & 0 & 0 & 0 & \dots & 0 \\ 0 & \lambda_2 & \dots & 0 & 0 & 0 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & 0 \\ 0 & 0 & \dots & \lambda_m & 0 & 0 & \dots & 0 \\ 0 & 0 & \dots & 0 & J_{\kappa_1} & 0 & \dots & 0 \\ 0 & 0 & \dots & 0 & 0 & J_{\kappa_2} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 0 & 0 & 0 & \dots & J_{\kappa_r} \end{bmatrix}.$$

In other words, every square matrix A is similar to a square matrix of the same dimension that is in Jordan canonical form.

8 [Optional material] Transforming to Jordan canonical form

Let $v_{\kappa_s,1}, v_{\kappa_s,2}, \dots, v_{\kappa_s, \mathfrak{g}_{\kappa_s}}$ be some *basis* of the null space of $A - \kappa_s I$. That is, $v_{\kappa_s,1}, v_{\kappa_s,2}, \dots, v_{\kappa_s, \mathfrak{g}_{\kappa_s}}$ are linearly independent and span all of the null space. Suppose that for each basis vector $v_{\kappa_s,j}$ there exist non-zero vectors $z_{\kappa_s,j,1}, z_{\kappa_s,j,2}, \dots, z_{\kappa_s,j,t_{s,j}}$, with $t_{s,j} \geq 1$ for $j = 1, \dots, \mathfrak{g}_{\kappa_s}$ and $t_{s,1} + t_{s,2} + \dots + t_{s, \mathfrak{g}_{\kappa_s}} = \mathfrak{a}_{\kappa_s} - \mathfrak{g}_{\kappa_s}$, such that

$$\begin{aligned} (A - \kappa_s I)z_{\kappa_s,j,1} &= v_{\kappa_s,j}, \\ (A - \kappa_s I)z_{\kappa_s,j,2} &= z_{\kappa_s,j,1}, \\ &\vdots \\ (A - \kappa_s I)z_{\kappa_s,j,t_{s,j}} &= z_{\kappa_s,j,t_{s,j}-1}. \end{aligned}$$

The sequence of vectors $v_{\kappa_s,j}, z_{\kappa_s,j,1}, z_{\kappa_s,j,2}, \dots, z_{\kappa_s,j,t_{s,j}}$ is referred to as a *Jordan chain* corresponding to the repeated eigenvalue κ_s . The vectors $z_{\kappa_s,j,1}, z_{\kappa_s,j,2}, \dots, z_{\kappa_s,j,t_{s,j}}$ are often referred to as *generalised eigenvectors*. All vectors in any one chain are linearly independent of one another and all vectors belonging to chains that belong to the same repeated eigenvalue but a different basis of the null space are linearly independent of one another. Also, the vectors in chains belonging to distinct repeated eigenvalues will be linearly independent of one another.

Define now the square matrix V_{κ_s} for $s = 1, 2, \dots, r$ as follows:

$$V_{\kappa_s} = \left[\begin{array}{c} \underbrace{v_{\kappa_s,1} \quad z_{\kappa_s,1,1} \quad \dots \quad z_{\kappa_s,1,t_{s,1}}}_{\text{Vectors of a Jordan chain}} \quad \underbrace{v_{\kappa_s,2} \quad z_{\kappa_s,2,1} \quad \dots \quad z_{\kappa_s,2,t_{s,2}}}_{\text{Vectors of a Jordan chain}} \quad \dots \\ \underbrace{v_{\kappa_s, \mathfrak{g}_{\kappa_s}} \quad z_{\kappa_s, \mathfrak{g}_{\kappa_s},1} \quad \dots \quad z_{\kappa_s, \mathfrak{g}_{\kappa_s}, t_{s, \mathfrak{g}_{\kappa_s}}}}_{\text{Vectors of a Jordan chain}} \end{array} \right].$$

Then by the construction of the Jordan chains given above we find that

$$AV_{\kappa_s} = V_{\kappa_s} \underbrace{\begin{bmatrix} J_{\kappa_s,1} & 0 & 0 & \dots & 0 \\ 0 & J_{\kappa_s,2} & 0 & \dots & 0 \\ 0 & 0 & J_{\kappa_s,3} & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & \dots & 0 & \dots & J_{\kappa_s, \mathfrak{g}_{\kappa_s}} \end{bmatrix}}_{J_{\kappa_s}},$$

where each $J_{\kappa_s,j}$ is a $(t_{s,j} + 1) \times (t_{s,j} + 1)$ Jordan block. They have the property that

$$\dim(J_{\kappa_s,1}) + \dim(J_{\kappa_s,2}) + \dots + \dim(J_{\kappa_s, \mathfrak{g}_{\kappa_s}}) = \mathfrak{a}_{\kappa_s}.$$

Now, let $u_1, u_2, \dots, u_m$ be eigenvectors corresponding to the non-repeating eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_m$ of A , respectively. These eigenvectors will be linearly independent of all the vectors in all of the Jordan chains of the repeated eigenvalues. Define the matrix V as

$$V = \begin{bmatrix} u_1 & u_2 & \dots & u_m & V_{\kappa_1} & V_{\kappa_2} & \dots & V_{\kappa_r} \end{bmatrix}.$$

Then by construction $AV = VJ$, where J is a matrix in Jordan canonical form, and therefore

$$V^{-1}AV = J.$$

In other words, such a matrix V transforms A into Jordan canonical form. What the Jordan canonical form J is has been implicitly given in the construction above, the sizes of the Jordan blocks depending on the integer quantities $t_{s,j}$ that still have to be determined.

For a given repeated eigenvalue κ_s with $\mathfrak{g}_{\kappa_s} \neq \mathfrak{a}_{\kappa_s}$, how do we find the Jordan blocks $J_{\kappa_s,1}, J_{\kappa_s,2}, \dots, J_{\kappa_s, \mathfrak{g}_{\kappa_s}}$ and V_{κ_s} ? This can be done algorithmically as follows¹:

1. Construct the null spaces $E_{\kappa_s}^k = \{z \in \mathbb{C} \mid (A - \kappa_s I)^k z = 0\}$ from $k = 1$ until $k = k(\kappa_s)$, where $k(\kappa_s)$ is the smallest integer such that $\dim(E_{\kappa_s}^{k(\kappa_s)}) = \mathfrak{a}_{\kappa_s}$.
2. Compute the numbers $d_1 = \dim(E_{\kappa_s}^1)$, $d_2 = \dim(E_{\kappa_s}^2) - \dim(E_{\kappa_s}^1)$, $d_3 = \dim(E_{\kappa_s}^3) - \dim(E_{\kappa_s}^2)$, $\dots$, $d_{k(\kappa_s)} = \dim(E_{\kappa_s}^{k(\kappa_s)}) - \dim(E_{\kappa_s}^{k(\kappa_s)-1})$.
3. Make $k(\kappa_s)$ rows of empty boxes with d_1 empty boxes in the first row, d_2 empty boxes in the second row, and so on until $d_{k(\kappa_s)}$ empty boxes in the $k(\kappa_s)$ -th row. For example, if $k(\kappa_s) = 4$, $d_1 = 4$, $d_2 = 3$, $d_3 = 2$ and $d_4 = 2$ then the rows of boxes will look like

$$\begin{array}{cccc} \square & \square & \square & \square \\ \square & \square & \square & \\ \square & \square & & \\ \square & \square & & \end{array}$$

¹The exposition here is based on http://math.rice.edu/~friedl/math355_fall04/Jordan.pdf

Starting at the last row $k = k(\kappa_s)$, fill the empty boxes in this row with d_k linearly independent vectors $v_{k,1}, v_{k,2}, \dots, v_{k,d_k}$ that is an element of $E_{\kappa_s}^k$ but **not** an element of $E_{\kappa_s}^{k-1}$. For every vector v that is placed in a box, fill the box immediately above it with the vector $(A - \kappa_s I)v$. If an empty box is at the bottom of its column and it is located in row i , fill this box with a vector that is in $E_{\kappa_s}^i$ but **not** in $E_{\kappa_s}^{i-1}$ and which is linearly independent of every other vector in the filled boxes in row i . Do this until all empty boxes are filled.

4. V_{κ_s} can be constructed as follows. Go through each column of the filled rows of boxes from the previous step, starting from the left column and from the top element of this column. The top element becomes the first column of V_{κ_s} , the next element in the column becomes the second column of V_{κ_s} , and so on until the bottom element of the column. Go to the next column of boxes, start at the top element and repeat, going column by column until all elements of every column has been placed as a column of V_{κ_s} .
5. The Jordan block $J_{\kappa_s, k}$ will be of a dimension that is determined by the number of boxes from Step 2 that is in column k . Thus, there are $d_1 = \mathfrak{g}_{\kappa_s}$ Jordan blocks, as expected.

Keep in mind that the above algorithm has to be applied to each repeated eigenvalue.

The next significant remark is now due. Jordan canonical forms are only useful for mathematical analysis and proving theorems. It cannot be used in numerical computations because it is in general impossible for numerical algorithms to distinguish repeated roots from several roots that are very close together, due to the finite precision arithmetic of digital computers. In general, Jordan canonical forms can only be determined analytically or by symbolic computer algebra (e.g., with Maple or Mathematica).

9 [Optional material] Examples - Jordan canonical form

To find the Jordan canonical form of a matrix A , the Jordan blocks, and transformation that brings A into its Jordan canonical form, we need only find the eigenvectors for all eigenvalues with algebraic multiplicity one and/or geometric multiplicity equal to its algebraic multiplicity, and apply the algorithm in the previous section for all remaining repeated eigenvalues with geometric multiplicity not equal to its algebraic multiplicity. This section provides some illustrative examples.

Example 16 Consider the matrix A given by

$$A = \begin{bmatrix} 1 & 1 & 2 & 1 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

The characteristic equation is $c_A(\lambda) = (\lambda - 1)^4$. Thus A has 1 as a single eigenvalue with $\mathbf{a}_1 = 4$. Let's examine what its geometric multiplicity is. We have that $A - I$ is

$$A - I = \begin{bmatrix} 0 & 1 & 2 & 1 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix},$$

and we see from this that the dimension of the null space of $A - I$ is 1 because its second, third and fourth columns are linearly independent. That is, $\mathbf{g}_1 = 1$. Therefore, the Jordan canonical form of A must only have a single Jordan block of size 4×4 . We use the algorithm of Section 8 to determine the Jordan block. Solving the equation $(A - I)v = 0$ we find that a solution is $v = (1, 0, 0, 0)^T$ (any other solution can in fact only be a scalar multiple this solution), so this is an eigenvector of A corresponding to the eigenvalue 1. For this eigenvalue, we have

$$(A - I)^2 = \begin{bmatrix} 0 & 0 & 2 & 2 \\ 0 & 0 & 0 & 2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}; (A - I)^3 = \begin{bmatrix} 0 & 0 & 0 & 2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}; (A - I)^4 = 0.$$

Since $\dim(\text{null}((A - I)^4)) = 4$, we have $k(1) = 4$. Also, we have that $E_1^1 = \text{span}\{(1, 0, 0, 0)^T\}$, $E_1^2 = \text{span}\{(1, 0, 0, 0)^T, (0, 1, 0, 0)^T\}$, $E_1^3 = \text{span}\{(1, 0, 0, 0)^T, (0, 1, 0, 0)^T, (0, 0, 1, 0)^T\}$, and $E_1^4 = \mathbb{C}^4$. Therefore, for this eigenvalue, $d_1 = d_2 = d_3 = d_4 = 1$, so we will have four rows and only a single column of boxes. For the bottom box we can choose the vector $v = (0, 0, 0, 1)^T$ since it is in E_1^4 but not in E_1^3 . Since $(A - I)v = (1, 0, 1, 0)^T$, $(A - I)^2v = (2, 2, 0, 0)^T$, and $(A - I)^3v = (2, 0, 0, 0)^T$. The filled boxes then look like

$$\begin{array}{c} (2, 0, 0, 0)^T \\ (2, 2, 0, 0)^T \\ (1, 0, 1, 0)^T \\ (0, 0, 0, 1)^T \end{array}$$

The corresponding Jordan block is

$$J_1 = V^{-1}AV = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix},$$

and the Jordan transformation matrix V is

$$V = \begin{bmatrix} 2 & 2 & 1 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

Example 17 Now, let

$$A = \begin{bmatrix} 1 & 1 & 1 & -2 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

The characteristic equation is $c_A(\lambda) = (\lambda - 1)^4$. Thus A has 1 as an eigenvalue with $\mathfrak{a}_1 = 4$. Here $A - I$ is

$$A - I = \begin{bmatrix} 0 & 1 & 1 & -2 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix},$$

which has two linearly independent columns. Therefore, the dimension of the null space is 2 and $\mathfrak{g}_1 = 2$. Also

$$(A - I)^2 = 0,$$

so that $k(1) = 2$ and $E_1^2 = \mathbb{C}^4$. It follows for this eigenvalue that $d_1 = 2$ and $d_2 = 2$, so there will be two rows of boxes with two boxes in each row. Two eigenvectors of A corresponding to the eigenvalue 1 are $v_{1,1} = (1, 0, 0, 0)^T$ and $v_{1,2} = (0, -1, 1, 0)^T$, so that $\text{null}(A - I) = \text{span}\{v_{1,1}, v_{1,2}\}$. So, the last row of boxes can be filled with the vectors $(0, 1, 0, 0)^T$ and $(0, 0, 0, 1)^T$ which are in E_1^2 but not in E_1^1 . The filled boxes are thus

$$\begin{array}{cc} (1, 0, 0, 0)^T & (-2, -1, 1, 0)^T \\ (0, 1, 0, 0)^T & (0, 0, 0, 1)^T \end{array}$$

The corresponding Jordan canonical form with two Jordan blocks is

$$J_1 = V^{-1}AV = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix},$$

and the Jordan transformation matrix V is

$$V = \begin{bmatrix} 1 & 0 & -2 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

10 Exponential of a matrix

Recall that for a scalar a (real or complex) we define the exponential e^a as the infinite sum

$$e^a = \sum_{k=0}^{\infty} \frac{1}{k!} a^k.$$

What if instead of a scalar a we have a square matrix A , can the exponential of a matrix A , e^A , be defined? Yes, it can be defined analogously to the scalar case. The exponential of a square matrix A is defined by the infinite sum

$$e^A = \sum_{k=0}^{\infty} \frac{1}{k!} A^k,$$

where I is an identity matrix of the same dimension as A .

So, how can we compute this infinite sum? To see how we do this, first consider the simplest case where A is a diagonal matrix $A = \text{diag}(a_1, a_2, \dots, a_n)$. It's straightforward to see that A^k is the diagonal matrix $A^k = \text{diag}(a_1^k, a_2^k, \dots, a_n^k)$, so that we have :

$$\begin{aligned} e^A &= \sum_{k=0}^{\infty} \frac{1}{k!} A^k, \\ &= \sum_{k=0}^{\infty} \frac{1}{k!} \text{diag}(a_1^k, a_2^k, \dots, a_n^k), \\ &= \text{diag} \left(\sum_{k=0}^{\infty} \frac{1}{k!} a_1^k, \sum_{k=0}^{\infty} \frac{1}{k!} a_2^k, \dots, \sum_{k=0}^{\infty} \frac{1}{k!} a_n^k \right), \\ &= \text{diag}(e^{a_1}, e^{a_2}, \dots, e^{a_n}). \end{aligned}$$

Thus, the exponential of a diagonal matrix A is simply a diagonal matrix with diagonal entries that are the exponentials of the corresponding diagonal entries of A .

Now, let's move on the next simplest case when A is diagonalisable (e.g., when A is symmetric). Then there exists an invertible square matrix V of the same dimension as A such $A = VDV^{-1}$ with $D = \text{diag}(d_1, d_2, \dots, d_n)$ a diagonal matrix, as we already saw. Now, note that:

$$A^k = \underbrace{AA \cdots A}_{k \text{ times}} = (VDV^{-1})(VDV^{-1}) \cdots (VDV^{-1}) = V \underbrace{DD \cdots D}_{k \text{ times}} V^{-1} = VD^k V^{-1}.$$

Therefore we have that

$$\begin{aligned} e^A &= \sum_{k=0}^{\infty} \frac{1}{k!} A^k, \\ &= \sum_{k=0}^{\infty} \frac{1}{k!} VD^k V^{-1}, \\ &= V \left(\sum_{k=0}^{\infty} \frac{1}{k!} D^k \right) V^{-1}, \\ &= Ve^D V^{-1}, \end{aligned}$$

where e^D is, of course, $e^D = \text{diag}(e^{d_1}, e^{d_2}, \dots, e^{d_n})$. So, to find the exponential of a diagonalisable matrix, first find a similarity transformation V^{-1} that diagonalises A into the

diagonal matrix D , take the exponential of D , and finally sandwich e^D with V^{-1} on the right and V on the left and take the matrix product of the three.

Now, to the general case where A is not necessarily diagonalisable. As you can already guess, Jordan blocks will replace the role of the diagonal matrix. To this end, there is an invertible square matrix V such that $A = VJV^{-1}$, where $J = \text{diag}(J_1, J_2, \dots, J_m)$ and $J_1, J_2, \dots, J_m$ are Jordan blocks. By the same calculations as before we find

$$e^A = Ve^JV^{-1},$$

where $e^J = \text{diag}(e^{J_1}, e^{J_2}, \dots, e^{J_m})$. Since J_k is not diagonal but is a bi-diagonal Jordan block, computing e^{J_k} is a bit more involved than the diagonal case. We will revisit this point and see how to determine e^{J_k} more conveniently in Week 5 using the Laplace transform (the basic idea has been introduced in ELEC3114), when we look at matrix exponentials of the form $e^{J_k t}$ for a time index $t \geq 0$ (in this case e^{J_k} is simply $e^{J_k t}$ evaluated at $t = 1$).

Some properties of the exponential of a matrix are:

1. If A and B commute (i.e., $AB = BA$) then $e^{A+B} = e^A e^B$ (not necessarily true if they don't commute!).
2. $(e^A)^{-1} = e^{-A}$ and therefore every exponential of a matrix is invertible.
3. If A is real then $(e^A)^T = e^{A^T}$.
4. If A is symmetric then e^{At} is symmetric and positive definite for all real t .

Example 18 Consider the matrix

$$A = \begin{bmatrix} -5 & 18 \\ -3 & 10 \end{bmatrix}.$$

This matrix has eigenvalues 1 and 4 and is diagonalisable, and is diagonalised by the matrix

$$V = \begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix}.$$

So, we have

$$\begin{aligned} e^A &= V \text{diag}(e, e^4) V^{-1}, \\ &= \begin{bmatrix} 3e - 2e^4 & -6e + 6e^4 \\ e - e^4 & -2e + 3e^4 \end{bmatrix}. \end{aligned}$$

11 Scalar fields, level sets, and gradients

- A scalar field on $\mathbb{R}^n$ is simply a real-valued function on $\mathbb{R}^n$. For example, $f(x, y, z) = x^3 + x^2 y z^2 + z^2$ is an example of a scalar field on $\mathbb{R}^3$.

- A c -level set of a scalar field f on $\mathbb{R}^n$ for a real number c is the set $V_c = \{x \in \mathbb{R}^n \mid f(x) = c\}$.
- The gradient of a scalar field f on $\mathbb{R}^n$, denoted by ∇f is a column vector defined as:

$$\nabla f(x) = \left(\frac{\partial f(x)}{\partial x_1}, \frac{\partial f(x)}{\partial x_2}, \dots, \frac{\partial f(x)}{\partial x_n} \right)^T.$$

- (A dot above a function x of time denotes the time derivative of the function)

$$\begin{aligned} \dot{f}(x(t)) &= \frac{d}{dt}f(x(t)) = \sum_{k=1}^n \frac{\partial f(x(t))}{\partial x_k} \dot{x}_k(t), \\ &= \nabla f(x(t))^T \dot{x}(t), \end{aligned}$$

where $\dot{x}(t) = (\dot{x}_1(t), \dot{x}_2(t), \dots, \dot{x}_n(t))^T$.

- Suppose that $x(t)$ is a differentiable trajectory lying on a c -level set, that is,

$$f(x(t)) = c.$$

Differentiating both sides gives $\nabla f(x(t))^T \dot{x}(t) = 0$. Now, $\dot{x}(t)$ is a vector that is tangent to the trajectory at the point $x(t)$, therefore $\nabla f(x)$ at a point x on a c -level set is a vector that points in a direction perpendicular to the tangent vector at the point x on the c -level set.

- Suppose that $\nabla f(x) \neq 0$. If f has a Taylor series, we can write

$$f(x + v) = f(x) + \nabla f(x)^T v + \text{higher order terms},$$

where the higher order terms are much smaller in magnitude than $|\nabla f(x)^T v|$ for sufficiently small $\|v\|$. Now, $-\|\nabla f(x)\|\|v\| \leq \nabla f(x)^T v \leq \|\nabla f(x)\|\|v\|$, and that $\nabla f(x)^T v = \|\nabla f(x)\|\|v\|$ whenever v points in the same direction as $\nabla f(x)$. With a detailed argument, it can be shown that this implies that $\nabla f(x)$ always points to a direction where $f(x)$ is increasing.